

Research Article

Hedwig Nora Nordlinder* and Jeremy Rubin

Gaussian Latent Averaging for scalar-on-Matrix regression

Subtitle do we need one?

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Abstract: Many biostatistical datasets contain different numbers of repeated observations for each individual (e.g. a varying number of histological samples per patient) that exhibit latent, within-individual heterogeneity. As an example, consider [what we want to consider]. A naive approach would be to model such unbalanced data would be to simply average the feature values across all subgroups within an individual. However, such averaging may lose valuable information about the heterogeneity between latent subgroups. An alternative strategy recently proposed is the CLUstering Structured lasSO (CLUSO) method which addresses this problem in the continuous-outcome setting by combining within-entity clustering with L^1 -penalised regression on the resulting empirical means of the clustered subgroup feature vectors. In this work, we propose a Bayesian generalisation of CLUSO for binary outcomes which we refer to as Gaussian Latent Averaging for scalar-on-Matrix regression (GLAM). Instead of committing to a single subgroup allocation for the computation of subgroup feature means, GLAM explicitly marginalises over the posterior predictive distribution of a Gaussian Mixture Model. GLAM is very flexible and allows a straight-forward generalisation to any GLM-problem including continuous, count and time-to-event outcomes. Further, using a Bayesian framework allows for more sophisticated sparsity-promotion compared to simple L^1 -regularisation as well as the incorporation of domain-specific knowledge to guide the relative prevalences of different subgroups. We demonstrate using simulated data that GLAM achieves a superior area under the precision-recall curve compared to classification done with naive averaging of entities across all subgroups per individual.

***Corresponding author: Hedwig Nora Nordlinder,** Department of Micro, Tumor and Cell Biology, Karolinska Institutet, Stockholm, Sweden

Jeremy Rubin, Department of Epidemiology and Biostatistics, University of Maryland School of Public Health, College Park, Maryland, USA

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1 Introduction

2 Method

2.1 Model definition

Blah blah blah we consider a model

2.1.1 Latent allocation model

Let x_{ij} denote the j :th observation of patient i . Let $\mathbf{1}_{ij}$ be the indicator for the observation belonging to subgroup 1. Let $\Sigma_{ik} = \lambda_{ik}^{-1} \Sigma_k$ be the covariance matrix for cluster k in patient i where Σ_k encodes the global correlation structure and λ_{ik} is a scalar variance correction to capture within-patient heterogeneity. Let μ_{ik} denote the patient-wise mean for cluster k . Let p_i be the probability of cluster 1 in each patient.

$$\begin{aligned} p_i &\sim \text{Beta}(\alpha_0, \beta_0) \\ z_{ik} &\sim \text{Ber}(p_i) \\ \mu_{ik} &\sim \mathcal{N}(\mu_{\text{hyper}}, \Lambda_0^{-1}) \\ m_i &\sim \mathcal{N}(0, I) \\ \lambda_{ik} &\sim \text{Gamma}(a_0, b_0) \\ \Sigma_k &\sim \mathcal{IW}(\nu_0, S_0) \\ x_{ij}|z_{ij} = k, \mu_{ik}, \lambda_{ik}, \Sigma_k &\sim \mathcal{N}(\mu_{ik}, \lambda_{ik}^{-1} \Sigma_k) \end{aligned}$$

2.1.2 Regression model

Let $\mathcal{Z}$ denote a sample from the allocation model. Conditional on $\mathcal{Z}$ let $X_i^{**} \in \mathbb{R}^{2 \times p}$ have as its first row the empirical mean of the feature vectors that by $\mathcal{Z}$ are allocated

to cluster 0 and let the second row be the same but for cluster 1. We then have some binary outcome y_i per patient and we run the regression

$$\begin{aligned} t &\sim \text{U}[0, 1] \\ \alpha &= (t, 1 - t) \\ \beta_i &\sim \mathcal{N}(0, 1) \\ p &\sim \text{Beta}(w_0, v_0) \\ \gamma_i &\sim \text{Ber}(p) \\ \text{logit } p_i &= \alpha^T X_i^{**} (\beta \odot \gamma) \\ y_i &\sim \text{Ber}(p_i) \end{aligned}$$

Here, we will use a Gibbs-Within-Langevin approach to sample the model. Specifically, we alternate between the continuous update

$$\beta_{k+1} = \beta_k + \frac{\tau}{2} \frac{\partial \ell(t, \beta_k, \gamma_k | \mathcal{X})}{\partial \beta} + \sqrt{\tau} \epsilon_{k+1}.$$

Now, let β_k^{active} be the coefficients satisfying $\gamma_k = 1$ and let $\beta_k^{\text{inactive}}$ be the coefficients satisfying $\gamma_k = 0$. We may decompose the update as

$$\beta_{k+1}^{\text{active}} = \beta_k^{\text{active}} + \frac{\tau}{2} \frac{\partial \ell(t, \beta_k^{\text{active}} | \mathcal{X})}{\partial \beta^{\text{active}}} + \sqrt{\tau} \epsilon_{k+1}$$

and

$$\begin{aligned} \beta_{k+1}^{\text{inactive}} &= \beta_k^{\text{inactive}} + \frac{\tau}{2} \frac{\partial \ell(t, \beta_k^{\text{inactive}} | \mathcal{X})}{\partial \beta^{\text{inactive}}} + \sqrt{\tau} \epsilon_{k+1} \\ &= \beta_k^{\text{inactive}} + \frac{\tau}{2} \frac{\partial (\ell(\mathcal{X} | t, \beta_k^{\text{inactive}}) + \ell(t, \beta_k^{\text{inactive}}))}{\partial \beta^{\text{inactive}}} + \sqrt{\tau} \epsilon_{k+1} \\ &= \beta_k^{\text{inactive}} + \frac{\tau}{2} \frac{\partial \ell(t, \beta_k^{\text{inactive}})}{\partial \beta^{\text{inactive}}} + \sqrt{\tau} \epsilon_{k+1} \quad (\text{Likelihood only depends on active parameters}) \\ &= \beta_k^{\text{inactive}} - \tau \beta_k^{\text{active}} + \sqrt{\tau} \epsilon_{k+1}. \end{aligned}$$

That is, the inactive parameters evolve under an AR(1)-process with mean 0 and mean reversion rate $(1 - \tau)$. The discrete update is given by

$$\gamma_{k+1}^{(j)} \sim \text{Ber}(p_\gamma^{(j)})$$

where

$$\begin{aligned} p_\gamma^{(j)} &:= \mathbb{P}[\gamma^{(j)} = 1 | \beta_k, \gamma_{k+1}^{(1)}, \dots, \gamma_{k+1}^{(j-1)}, \gamma_k^{(j+1)}, \dots, \gamma_k^{(p)}, t_k, p, \mathcal{X}] \\ &= \text{expit} \left(\frac{p}{1-p} + \ell_1 - \ell_0 \right) \end{aligned}$$

where ℓ_1 is the log-likelihood of the model when $\gamma^{(j)}$ is one and ℓ_0 is the log-likelihood of the model when $\gamma^{(j)}$ is zero

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